Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A compound 'A' when treated with HNO_3 (in presence of H_2SO_4) gives compound 'B', which is then reduced with Sn and HCl to aniline? The compound ' A ' is

(A) toluene

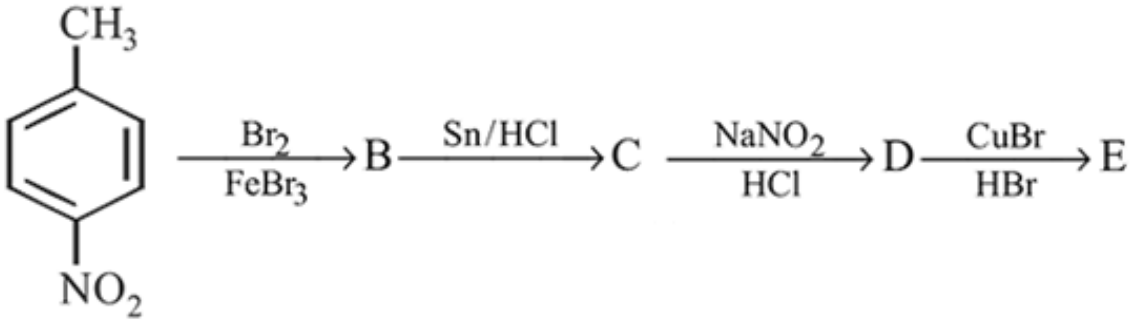
(B) benzene

(C) ethane

(D) acetamide

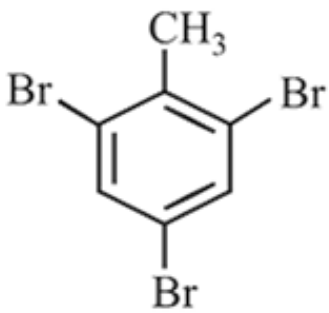
Q.2

In a set of reactions p -nitrotoluene yielded a product E.

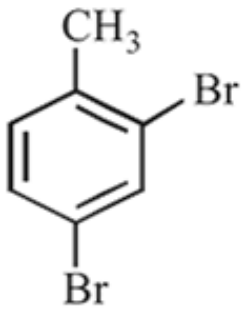


The product E would be:

(A)

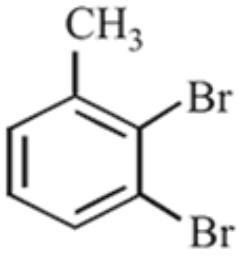


(B)

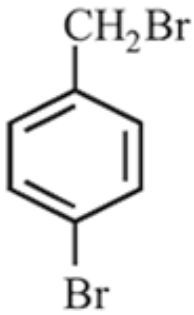


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(C)



(D)



Q.3

In the following reaction, X is

$$X \xrightarrow{\text{Bromination}} Y \xrightarrow{NaNO_2/HCl} Z \xrightarrow[C_2H_5OH]{\text{Boiling}} \text{Tribromobenzene}$$

(A) benzoic acid

(B) salicylic acid

(C) phenol

(D) aniline

Q.4

Aniline is reacted with bromine water and the resulting product is treated with an aqueous solution of sodium nitrite in presence of dilute HCl. The compound so formed is converted into tetrafluoroborate which is subsequently heated dry. The final product is

(A) p -Bromofluorobenzene

(B) p -Bromoaniline

(C) $2, 4, 6$ -Tribromofluorobenzene

(D) $1, 3, 5$ -Tribromobenzene

Q.5

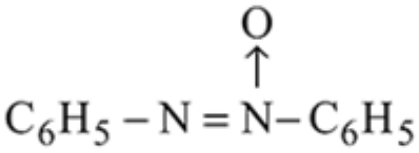
In the acidic reduction of nitrobenzene, which of the following is the intermediate?

(A) $C_6H_5 - N = O$

(B) $C_6H_5NH - NHC_6H_5$

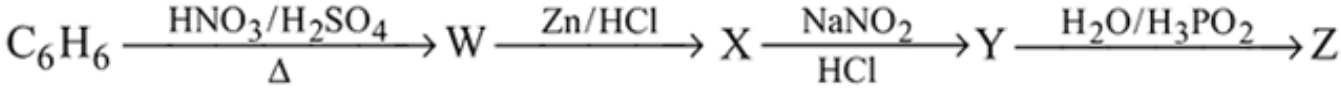
(C) $C_6H_5 - N = N - C_6H_5$

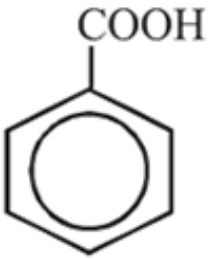
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(D) 

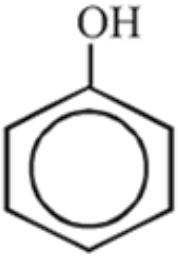
Q.6

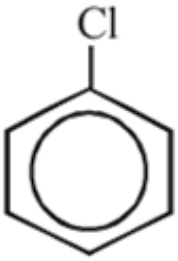
'Z' in the following sequence of reactions is



(A) 

(B) 

(C) 

(D) 

Q.7

Which of the following can undergo electrophilic substitution when treated with nitrous acid at $0^{\circ}C$?

(A) $C_6H_5NH_2$

(B) $C_6H_5NHCH_3$

(C) $C_6H_5\ N\ (CH_3)_2$

(D) None

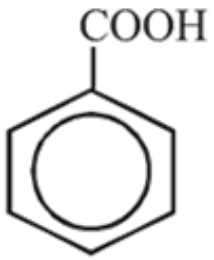
Q.8

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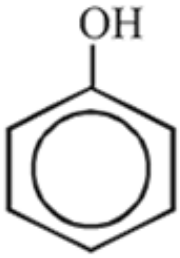
'Z' in the following sequence of reactions is

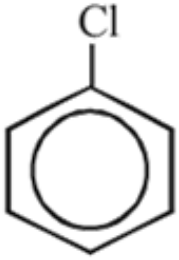
$$C_6H_6 \xrightarrow[\Delta]{HNO_3/H_2SO_4} W \xrightarrow{Zn/HCl}$$

$$X \xrightarrow[HCl]{NaNO_2} Y \xrightarrow{H_2O/H_3PO_2} Z$$

(A) 

(B) 

(C) 

(D) 

Q.9

Which of the following amines would undergo diazotisation

(A) Primary aliphatic amines

(B) Primary aromatic amines

(C) Both (a) and (b)

(D) None of these

Q.10

Which of the following reaction can produce $R - CO - Ar$?

(A) $ArCOCl + H - Ar \xrightarrow{AlCl_3}$

(B) $RCOCl + ArMgX \longrightarrow$

(C) $ArCOCl + RMgX \longrightarrow$

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(D) $\mathrm{RCOCl + H - Ar \xrightarrow{AlCl_3}}$

Q.11

Which of the following reaction can produce $\mathrm{R - CO - Ar?}$

(A) $\mathrm{ArCOCl + H - Ar \xrightarrow{AlCl_3}}$

(B) $\mathrm{RCOCl + ArMgX \longrightarrow}$

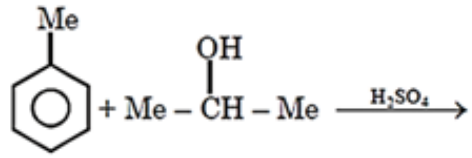
(C) $\mathrm{ArCOCl + RMgX \longrightarrow}$

(D) $\mathrm{RCOCl + H - Ar \xrightarrow{AlCl_3}}$

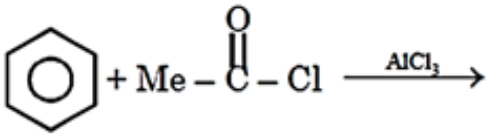
Q.12

How many reaction are named as Friedel Craft's reaction:

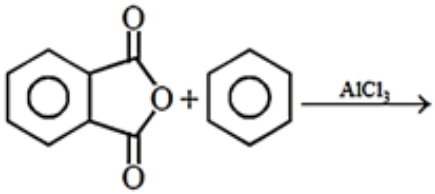
A



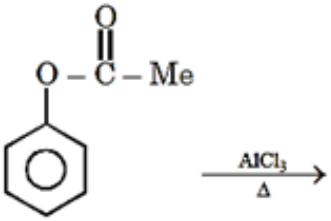
(B)



C



D



(A) 2.00

(B) 1.00

(C) 5.00

(D) 3.00

Q.13

When isobutene reacts with benzene, which of the following product will be obtained?

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(A) Isobutyl benzene

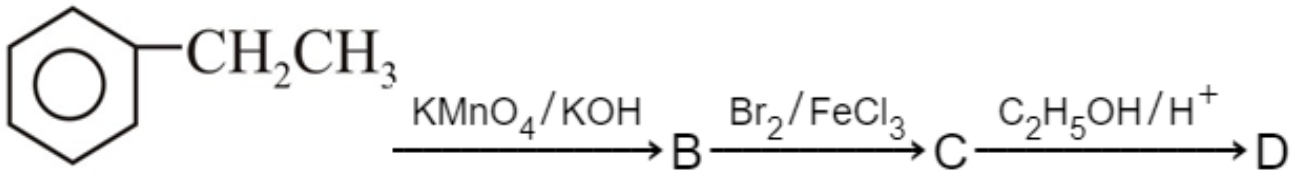
(B) Tert-butyl benzene

(C) n-butyl benzene

(D) No reaction

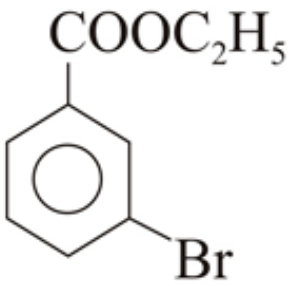
Q.14

In a set of reactions, ethyl benzene yielded a product D

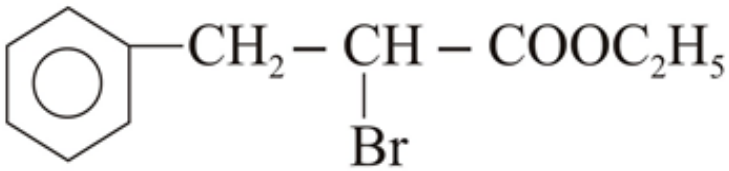


'D' Would be :-

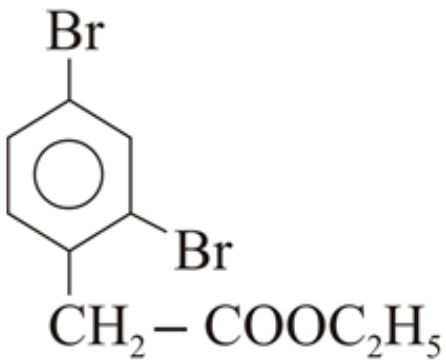
(A)



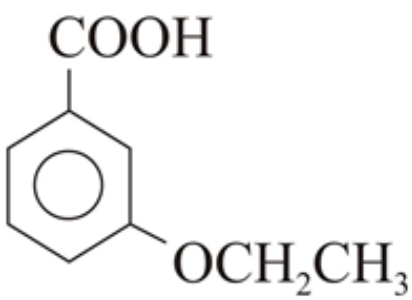
(B)



(C)



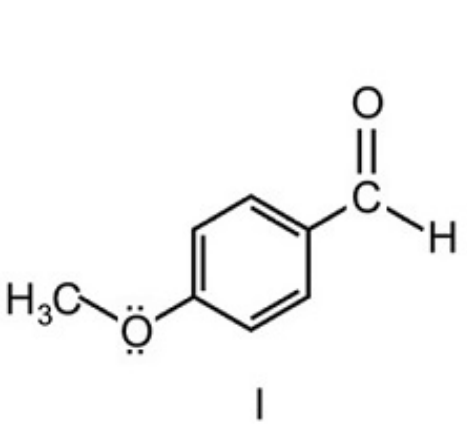
(D)



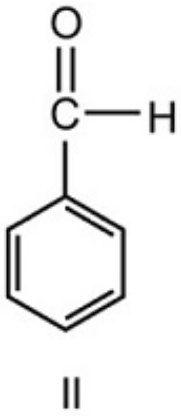
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Q.15

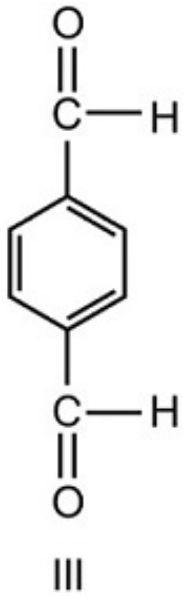
The K_{eq} value order for the addition of HCN, to the following is:



I



II



III

(A) $\mathrm{III} > \mathrm{II} > \mathrm{I}$

(B) $\mathrm{II} > \mathrm{I} > \mathrm{III}$

(C) $\mathrm{I} > \mathrm{II} > \mathrm{III}$

(D) $\mathrm{I} > \mathrm{III} > \mathrm{II}$

Answers & Solutions

Q.1 Answer:

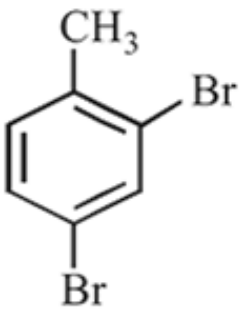
benzene

Solution:

$$\mathrm{A} \xrightarrow{\mathrm{HNO_3/H_2SO_4}} \mathrm{B} \xrightarrow{\mathrm{Sn/HCl}} \mathrm{C_6H_5NH_2}$$

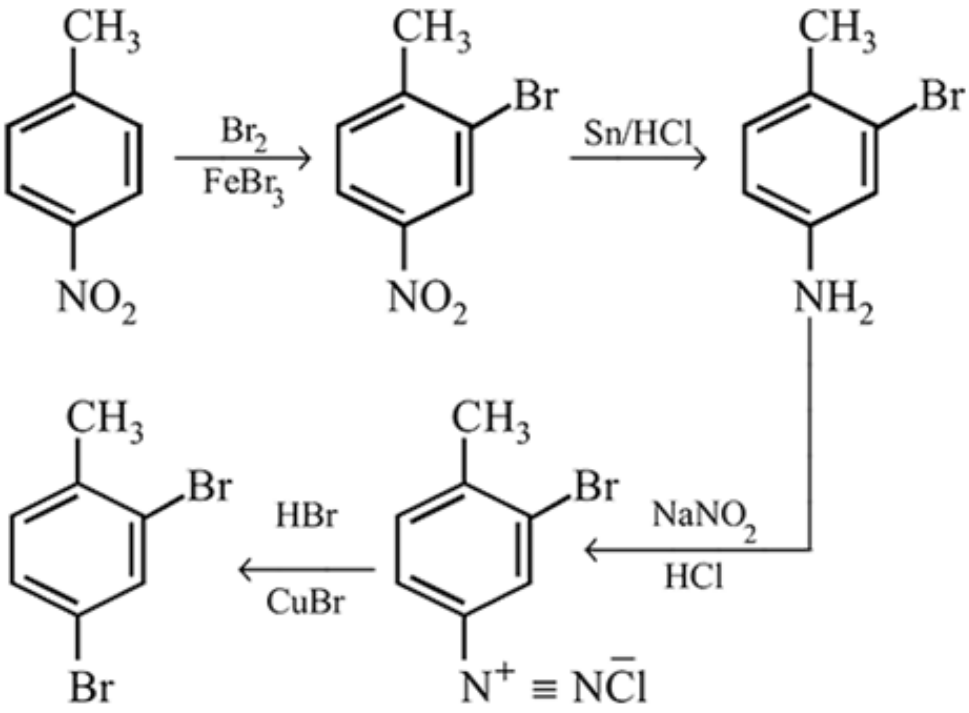
This indicates that B **is** $\mathrm{C_6H_5NO_2}$ **and hence** A **is** $\mathrm{C_6H_6}$

Q.2 Answer:



Solution:

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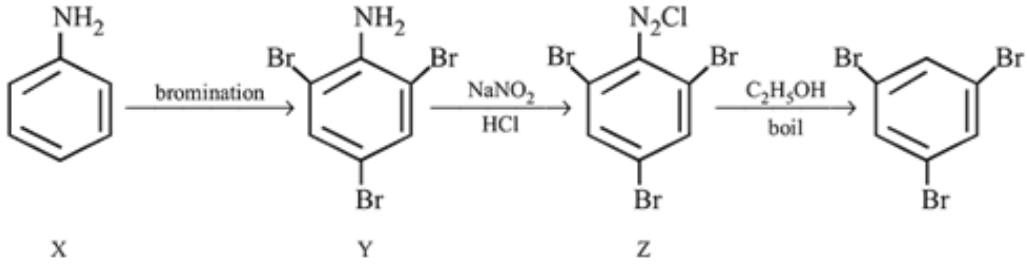


Q.3 Answer:

aniline

Solution:

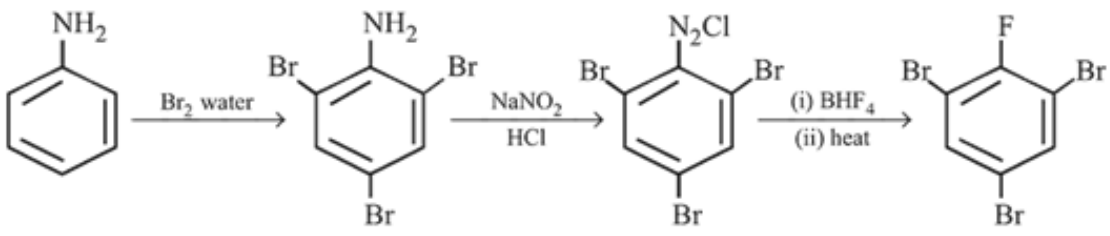
Proceed backward ; tribromobenzene is produced by boiling compound Z with C_2H_5OH; Z in turn 15 obtained by diazotisation of Y, so Y and Z should have $-NH_2$ and $-N_2Cl$ groups respectively, in addition to three Br atoms. Hence X should be $C_6H_5NH_2$



Q.4 Answer:

2, 4, 6 -Tribromofluorobenzene

Solution:



Q.5 Answer:

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$C_6H_5 - N = O$

Solution:

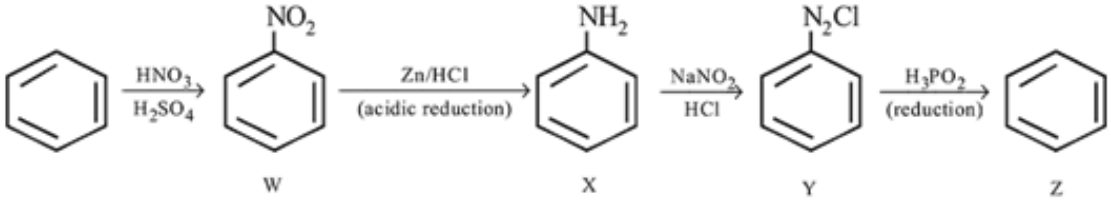
Nitrobenzene is reduced in the following manner through various intermediates to form aniline as the final product.

$$\underset{\text{nitrobenzene}}{C_6H_5NO_2} \xrightarrow{2[H]} \underset{\text{nitrosobenzene}}{C_6H_5NO} \xrightarrow{2[H]} \underset{\text{phenylhydroxylamine}}{C_6H_5NHOH} \xrightarrow{2[H]} \underset{\text{aniline}}{C_6H_5NH_2}$$

Q.6 Answer:



Solution:

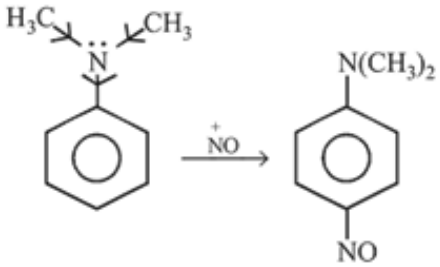


Q.7 Answer:

$C_6H_5\,N\,(CH_3)_2$

Solution:

Nitrosonium ion, $\overset{+}{N}O$ from HONO is a weak electrophile, hence it can attack only on highly activated benzene nucleus, provided proper position, *p* - or *o* - is free.

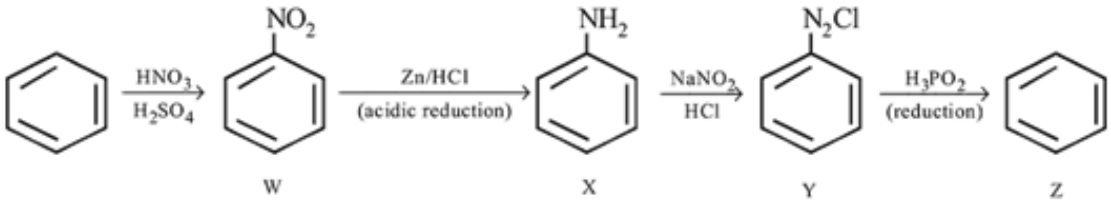


Q.8 Answer:



Solution:

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Q.9 Answer:

Primary aromatic amines

Solution:

Only primary aromatic amines can undergo diazotisation

Q.10 Answer:

$\mathrm{RCOCl} + \mathrm{H} - \mathrm{Ar} \xrightarrow{\mathrm{AlCl_3}}$

Solution:

ArCOR can be prepared by the combination of $\mathrm{ArH} + \mathrm{RCOCl}$ and not by $\mathrm{ArCOCl} + \mathrm{RMgX}$ because here the ArCOR formed will further react with RMgX to form 3° alcohol, $\mathrm{ArC(OH)R_2}$ as the final products.

Q.11 Answer:

$\mathrm{RCOCl} + \mathrm{H} - \mathrm{Ar} \xrightarrow{\mathrm{AlCl_3}}$

Solution:

ArCOR can be prepared by the combination of $\mathrm{ArH} + \mathrm{RCOCl}$ and not by $\mathrm{ArCOCl} + \mathrm{RMgX}$ because here the ArCOR formed will further react with RMgX to form 3° alcohol, $\mathrm{ArC(OH)R_2}$ as the final products.

Q.12 Answer:

3.00

Solution:

B, C and D.

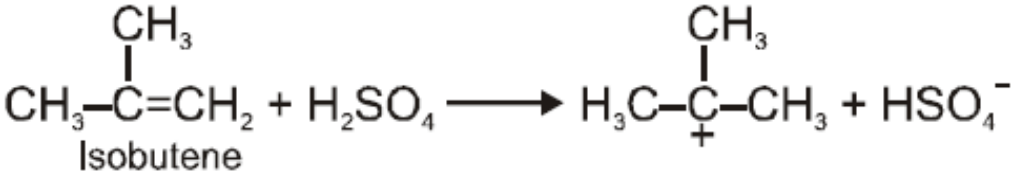
Q.13 Answer:

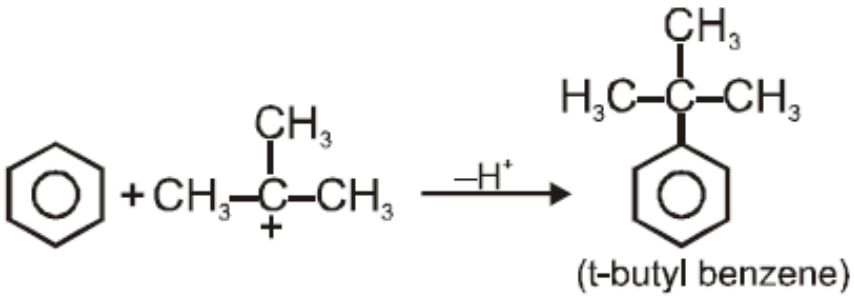
Tert-butyl benzene

Solution:

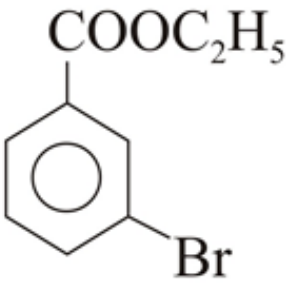
Benzene reacts with isobutene in the presence of $\mathrm{H_2SO_4}$ to produce t-butyl benzene. The reaction takes place as follows.

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Q.14 Answer:



Solution:

...

Q.15 Answer:

III > II > I

Solution:

More +ve charge on carbonyl compound C-atom; more reactive towards nucleophilic addition

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